MA1522 Linear Algebra for Computing

Lesson 7: Euclidean Spaces, Dot Product, and Linear Spans

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson7` (e.g. `A1234567Z_lesson7.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 7**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Basic rules:

- Please (mainly) ask the questions related to the course.

Section 3.1 Euclidean Vector Spaces

The **Euclidean n-space** $\mathbb{R}^n$ is the collection of all n-vectors $\mathbf{v} = (v_i)$, where $v_i \in \mathbb{R}$ is the i-th **coordinate**. A vector is drawn as an arrow with its tail at the origin, or as the point at its head.

A set V with an addition and a scalar multiplication is a **vector space** over $\mathbb{R}$ if it satisfies these ten axioms. Here $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are in V and $a, b \in \mathbb{R}$.

1. $\mathbf{u} + \mathbf{v}$ is in V;
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$;
4. there is $\mathbf{0}$ in V with $\mathbf{0} + \mathbf{v} = \mathbf{v}$;
5. each $\mathbf{u}$ has $-\mathbf{u}$ in V with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
6. $a\mathbf{v}$ is in V;
7. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
8. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$;
9. $a(b\mathbf{u}) = (ab)\mathbf{u}$;
10. $1\mathbf{u} = \mathbf{u}$.

Since vectors are matrices, matrix algebra gives all ten for $\mathbb{R}^n$.

Section 3.2 Dot Product, Norm, Distance

The **dot product** (or **inner product**) of $\mathbf{u} = (u_i)$ and $\mathbf{v} = (v_i)$ in $\mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

Norm, unit vector

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + \cdots + u_n^2}.$$

$\mathbf{u}$ is a **unit vector** if $\|\mathbf{u}\| = 1$; for $\mathbf{u} \neq \mathbf{0}$, **normalizing** gives one: $\mathbf{u}/\|\mathbf{u}\|$.

Distance, angle

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the **angle** θ between them, with $0 \leq \theta \leq \pi$, is given by $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$.

The dot product is symmetric and distributes over sums, and $\mathbf{u} \cdot \mathbf{u} \geq 0$ with equality only for $\mathbf{u} = \mathbf{0}$. Scalars come out of both: $\mathbf{u} \cdot (c\mathbf{v}) = c\,(\mathbf{u} \cdot \mathbf{v})$ and $\|c\mathbf{u}\| = |c| \, \|\mathbf{u}\|$.

Section 3.3 Linear Combinations and Linear Spans

Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be vectors in $\mathbb{R}^n$. A **linear combination** of them is $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k$, where the scalars $c_1, \ldots, c_k \in \mathbb{R}$ are the **coefficients**. Their **span** is the set of all linear combinations,

$$\operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} = \{\, c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k \mid c_1, \ldots, c_k \in \mathbb{R} \,\} \subseteq \mathbb{R}^n.$$

Algorithm: is $\mathbf{v}$ a linear combination?

Form the $n \times k$ matrix $\begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$ with the vectors as its columns. Then

$$\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \iff \left(\begin{array}{ccc|c} \mathbf{u}_1 & \cdots & \mathbf{u}_k & \mathbf{v} \end{array} \right) \text{ is consistent,}$$

and the solutions of that system are exactly the possible coefficients.

Question 1

Let

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad \mathbf{w}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{w}_3 = \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}.$$

1. Is $\mathbf{v}$ in $\operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$? If it is, write $\mathbf{v}$ as a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.

2. Are the coefficients unique? If not, give two different sets of coefficients.

Question 1 — solution: part (1)

Put the three vectors in the columns of an augmented matrix and reduce.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 5 \\ 3 & 0 & -3 & 6 \end{array}\right) \xrightarrow[R_3-3R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -3 \\ 0 & -6 & -6 & -6 \end{array}\right) \xrightarrow[-\frac{1}{3}R_2]{R_3-2R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_1-2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The last row is $0 = 0$, so the system is consistent and $\mathbf{v} \in \operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$. Reading off the solution with $c_3 = 0$ gives $c_1 = 2$, $c_2 = 1$, so

$$\mathbf{v} = 2\mathbf{w}_1 + \mathbf{w}_2 = 2\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}. \checkmark$$

Question 1 — solution: part (2)

Not unique. The reduced system has c_3 as a free variable, so the general solution is

$$c_1 = 2 + t, \qquad c_2 = 1 - t, \qquad c_3 = t, \qquad t \in \mathbb{R}.$$

Taking $t = 0$ and $t = 1$ gives two different sets of coefficients:

$$\mathbf{v} = 2\mathbf{w}_1 + 1\mathbf{w}_2 + 0\mathbf{w}_3 \qquad \text{and} \qquad \mathbf{v} = 3\mathbf{w}_1 + 0\mathbf{w}_2 + 1\mathbf{w}_3.$$

Check the second: $3(1,2,3)^T + (1,-1,-3)^T = (4,5,6)^T$. ✓

The two differ because $\mathbf{w}_3 = \mathbf{w}_2 - \mathbf{w}_1$, as you can check directly: one of the three vectors was never needed. □

Section 3.3 When is $\text{span}(S) = \mathbb{R}^n$?

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\} \subseteq \mathbb{R}^n$. View the vectors as column vectors, and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then

$$
\begin{aligned}
\text{span}(S) = \mathbb{R}^n &\iff \text{for all } \mathbf{v} \in \mathbb{R}^n,\ A\mathbf{x} = \mathbf{v} \text{ has a solution} \\
&\iff \text{for all } \mathbf{v} \in \mathbb{R}^n,\ A\mathbf{x} = \mathbf{v} \text{ is consistent} \\
&\iff \text{RREF}(A) \text{ has no zero row} \\
&\iff \text{all rows of RREF}(A) \text{ are nonzero} \\
&\iff \text{all rows of any row-echelon form of } A \text{ are nonzero.}
\end{aligned}
$$

Example 1

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of k vectors in $\mathbb{R}^n$. True or false?

1. If $k < n$, then $\text{span}(S) \neq \mathbb{R}^n$.
2. If $k \geq n$, then $\text{span}(S) = \mathbb{R}^n$.

Explain your answer to your neighbour.

(1) **True.** Let R be the reduced row-echelon form of the $n \times k$ matrix $A = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$. Each nonzero row of R carries one leading entry, and each leading entry sits in a column of its own, so

$$\#\{\text{nonzero rows of } R\} \leq \#\{\text{columns of } R\} = k < n = \#\{\text{rows of } R\}.$$

So R has a zero row, and $\text{span}(S) \neq \mathbb{R}^n$.

(2) **False.** Counting vectors is not enough. In $\mathbb{R}^2$ take $k = 2$ with $\mathbf{u}_1 = \mathbf{u}_2 = (1, 0)^T$: the span is one line, not $\mathbb{R}^2$.

Question 2

Let

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}.$$

1. Show that $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \mathbb{R}^3$, and write each of $\mathbf{v}$ and $\mathbf{w}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

2. For $i = 1, 2, 3$, compute $\dfrac{\mathbf{v} \cdot \mathbf{u}_i}{\|\mathbf{u}_i\|^2}$ and $\dfrac{\mathbf{w} \cdot \mathbf{u}_i}{\|\mathbf{u}_i\|^2}$, and compare them with your coefficients from part (1). What do you notice?

Question 2 — solution: part (1)

Both vectors share the same coefficient matrix, so reduce once with two columns on the right.

$$A = \left(\begin{array}{ccc|cc} 1 & 2 & 2 & 3 & 2 \\ 2 & 1 & -2 & 6 & 1 \\ 2 & -2 & 1 & -3 & 7 \end{array}\right), \qquad \mathrm{RREF}(A) = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right).$$

The left-hand block has no zero row, so $\mathrm{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \mathbb{R}^3$.

The two right-hand columns are the coefficients:

$$\mathbf{v} = \mathbf{u}_1 + 2\mathbf{u}_2 - \mathbf{u}_3, \qquad \mathbf{w} = 2\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3.$$

Check the second: $2(1,2,2)^T - (2,1,-2)^T + (2,-2,1)^T = (2,1,7)^T$. ✓

Question 2 — solution: part (2)

Each denominator is a vector dotted with itself:

$$\|\mathbf{u}_1\|^2 = 1+4+4 = 9, \qquad \|\mathbf{u}_2\|^2 = 4+1+4 = 9, \qquad \|\mathbf{u}_3\|^2 = 4+4+1 = 9.$$

	$i=1$	$i=2$	$i=3$
$\dfrac{\mathbf{v}\cdot\mathbf{u}_i}{\|\mathbf{u}_i\|^2}$	$\dfrac{3+12-6}{9} = 1$	$\dfrac{6+6+6}{9} = 2$	$\dfrac{6-12-3}{9} = -1$
$\dfrac{\mathbf{w}\cdot\mathbf{u}_i}{\|\mathbf{u}_i\|^2}$	$\dfrac{2+2+14}{9} = 2$	$\dfrac{4+1-14}{9} = -1$	$\dfrac{4-2+7}{9} = 1$

Those are exactly the coefficients of $\mathbf{v}$ and of $\mathbf{w}$ from part (1). For these three vectors the coefficients of any vector can be read off this way, with no reduction at all. □

Question 3

Let

$$S_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}, \qquad S_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

1. Determine whether each of S_1 and S_2 spans $\mathbb{R}^3$.

2. For the set that does not, find a vector of $\mathbb{R}^3$ that is not in its span.

Question 3 — solution: part (1)

Put each set in the columns of a matrix and reduce. Two row operations each.

S_1:

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow[R_3 - R_2]{R_2 - R_1} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

No zero row, so $\operatorname{span}(S_1) = \mathbb{R}^3$.

S_2:

$$\begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 - R_2]{R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

A zero row, so $\operatorname{span}(S_2) \neq \mathbb{R}^3$.

S_2 has four vectors and S_1 has three, so the number of vectors settles nothing.

Question 3 — solution: part (2)

A vector $\mathbf{v} = (x, y, z)^T$ is in $\operatorname{span}(S_2)$ exactly when the system is consistent, so carry $\mathbf{v}$ through the same two row operations as an extra column.

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 1 & 1 & 2 & 1 & z \end{array}\right) \xrightarrow[R_3 - R_2]{R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 0 & 0 & 0 & 0 & z - x - y \end{array}\right).$$

The last row reads $0 = z - x - y$, so the system is consistent exactly when $z = x + y$. Any vector failing that is outside the span; for instance

$$\mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad z - x - y = 1 \neq 0.$$

So $\operatorname{span}(S_2)$ is the plane $z = x + y$, and $(0, 0, 1)^T$ is not on it. □

Next lesson

- **Sections 3.3 & 3.4 Linear Spans and Subspaces**

Please watch the video clips for these sections beforehand.